

Project Presentation of IOHE

On

Smart Campus

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1.1 Background

- **Manual Management:** Seminar halls and study rooms are managed manually without a proper system.
- **Availability Problems:** Students do not know if a room is available in real time, which causes wasted time and booking clashes.
- **Poor Resource Usage:** Some users keep spaces occupied for too long because there are no time limits.
- **Lack of Monitoring:** There is no central system to track room usage or make sure all students get fair access to facilities.

1.2 Problem Statement

- **Centralized Management:** Create one digital system to easily book and manage shared college spaces.
- **Role-Based Governance:** Use roles like “Community Head” and “Community Member” to manage activities and responsibilities properly.
- **Fairness & Transparency:** Improve campus management by giving real-time updates and proper monitoring so everyone gets fair access to resources.

The main **objectives** of Smart Campus are to improve the management and utilization of shared university spaces through a smart and centralized digital platform.

1. **Digital Management:** Develop a digital system for booking and managing shared university spaces efficiently.
2. **Role-Based Access:** Organize users through roles such as Community Head and Community Member for better management and coordination.
3. **Fair Resource Allocation:** Provide fair and transparent access to seminar halls and study rooms for all users.
4. **Efficient Monitoring:** Improve the monitoring and utilization of university resources through centralized management.
5. **Time-Based Usage Control:** Set proper time limits for room usage to avoid unnecessary occupation and ensure better availability.

Proposed Solution



The proposed system is a smart digital platform designed to manage shared university spaces such as seminar halls, study rooms, and meeting rooms in an easy and organized way. The system helps students and administrators book, monitor, and manage spaces efficiently without manual work.

Main Functions of the System:-

- **Online Room Booking:** Students and faculty can book seminar halls or study rooms online through the system without visiting the office physically.
- **Real-Time Availability Checking:** Users can instantly check whether a room is available or already booked, which helps avoid confusion and scheduling conflicts.
- **Time-Based Booking System:** The system allows bookings for a limited time period so that resources are used fairly and no one occupies a room for too long.
- **Role-Based Access:** Different roles such as Community Head and Community Member are provided to manage activities in a structured way. Community Heads can monitor and approve bookings if required.

Project Structure



Smart Campus Booking System - Project Structure

Backend (Node.js + Express)

- server.js - Main entry point
- config.env - Environment vars
- **controllers/** - Business logic
 - └ authController.js
 - └ userController.js
 - └ bookingController.js
 - └ eventController.js
 - └ aiController.js
- **models/** - DB schemas
 - └ User.js
 - └ Event.js
 - └ Booking.js
- **routes/** - API endpoints
 - └ auth.js, user.js, booking.js
 - └ event.js, ai.js
- **middleware/** - Auth & error handling
- **utils/** - Helpers & services

Frontend (React + TypeScript)

- App.tsx - Main component
- index.tsx - Entry point
- **context/** - State management
 - └ AuthContext.tsx
- **components/** - Reusable UI
 - └ Navbar.tsx
 - └ SeatMap.tsx
 - └ EventSeatMap.tsx
 - └ SmartSearch.tsx
 - └ SmartAssistant.tsx
- **pages/** - Route pages (12 pages)
 - └ Home, Login, Signup
 - └ Dashboard, MyBookings
 - └ CreateEvent, AdminPanel
 - └ ForgotPassword, etc.

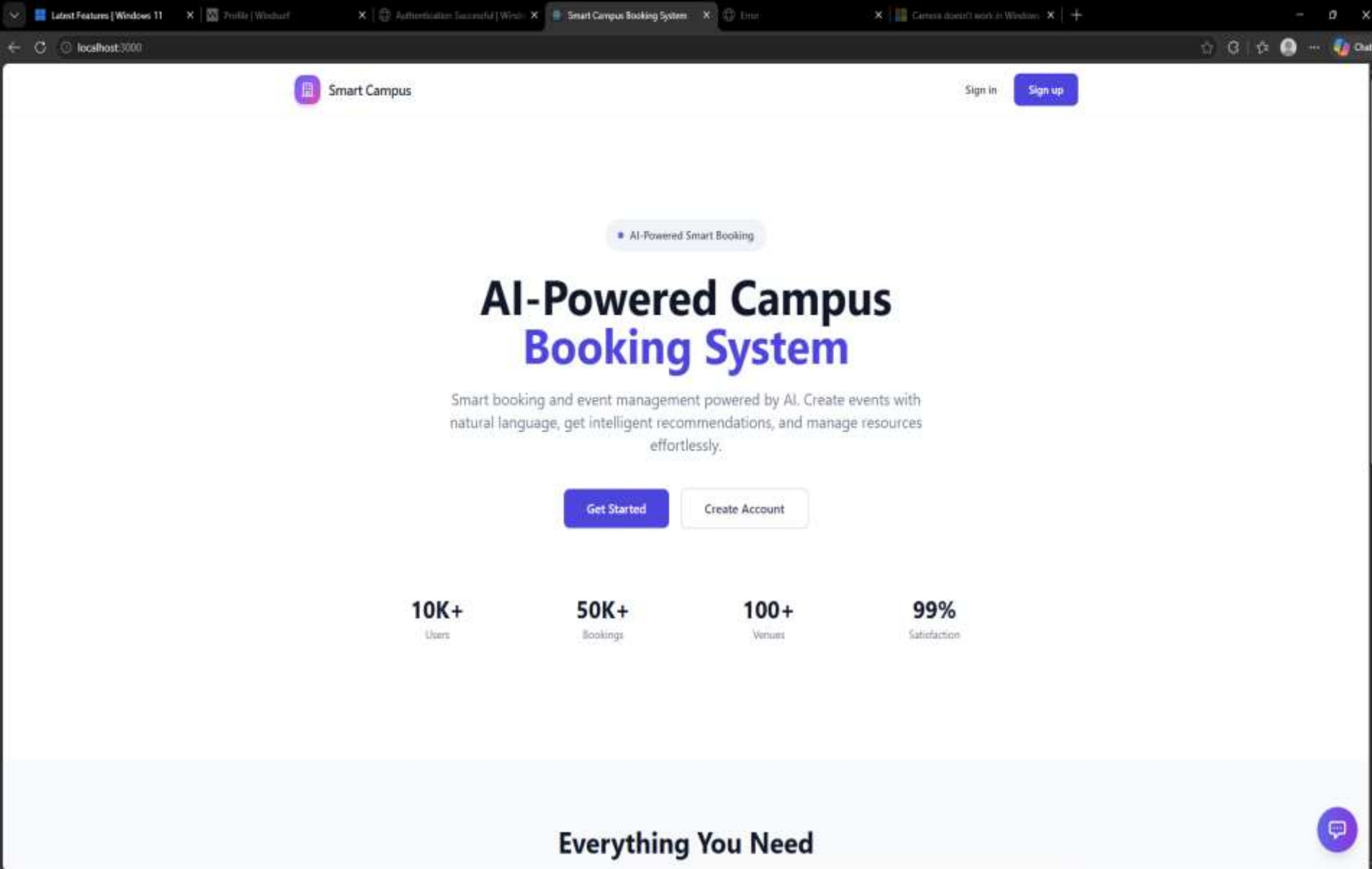
Smart Campus is built using the **MERN Stack**, which provides a scalable, secure, and centralized platform for managing shared university resources efficiently.

Technologies:-

- **React.js (with Vite):** Used to create a fast, dynamic, and user-friendly interface.
- **Node.js & Express.js:** Handles the server-side operations of the application.
- **MongoDB:** A NoSQL database used to store application data efficiently.
- **JWT (JSON Web Tokens):** Used for secure user authentication and authorization.
- **Tailwind CSS:** Used for designing a clean, responsive, and professional interface.
- **REST APIs:** Enable smooth communication between the frontend and backend.

VISUAL REPRESENTATION

CHITKARA
UNIVERSITY



VISUAL REPRESENTATION



Browser tabs: Profile | Wintur | Authentication Successful | Wintur | New tab | Smart Campus Booking System

URL: localhost:3000/dashboard

Smart Campus

Home | Dashboard | Booking | Create Event | Join Event | Admin Panel

Abhishek | Sign out

A

Welcome back, Abhishek

+ New Booking

Administrate

Ready to book your next space

TOTAL BOOKINGS

2

+12%

APPROVED

0

+8%

EVENTS JOINED

0

+15%

PENDING

0

-3%

Quick Actions

New Booking

Click to proceed

Create Event

Click to proceed

Join Event

Click to proceed

Recent Bookings (2)

View all >

Seminar Hall completed

Thu, May 7 at 09:00

Seats: 28

Canteen completed

Fri, May 8 at 16:00

Seats: 47

Upcoming Events (0)

View all >

No events yet

Create Event

Join an existing event

VISUAL REPRESENTATION

CHITKARA
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Profile | Windsurf Authentication Successful | Windsurf New tab Smart Campus Booking System

localhost:3000/admin

Smart Campus Home Dashboard Booking Create Event Join Event Admin Panel Abhishek Sign out

Manage bookings, users, and events

Bookings (10) Users (0) Events (0)

Booking Requests

USER	RESOURCE	DATE & TIME	PURPOSE	STATUS	ACTIONS
Abhay Kumar Singh abhay1146.be22@chitkara.edu.in	Canteen	8/5/2026 at 00:30	good	completed	
Abhishek abhishek9097indian@gmail.com	Seminar Hall	7/5/2026 at 00:00	this is good	completed	
abhishek kumar abhishek1167.be22@chitkara.edu.in	Lecture Hall	9/5/2026 at 14:00	sksk	completed	
abhishek kumar abhishek1167.be22@chitkara.edu.in	Seminar Hall	9/5/2026 at 17:00	this is good	completed	
Abhishek abhishek9097indian@gmail.com	Canteen	8/5/2026 at 16:00	this is fun	completed	
Unknown User N/A	Canteen	8/5/2026 at 12:00	sdfghj	completed	
Unknown User N/A	Seminar Hall	7/5/2026 at 10:00		completed	
Unknown User N/A	Canteen	9/5/2026 at 11:00	...	completed	
Unknown User N/A	Canteen	16/5/2026 at 12:00	good things	approved	
Unknown User N/A	Seminar Hall	9/5/2026 at 12:00	xyz	completed	

VISUAL REPRESENTATION



Profile | Windsurf Authentication Successful | Windsurf New tab Smart Campus Booking System

localhost:3000/join-event

Smart Campus Home Dashboard Booking Create Event Join Event Admin Panel Abhishek Sign out

Join Event

Enter the event details to join an existing event

Event Name

Enter the event name

Room Code

ENTER 6-CHARACTER CODE

Find Event & Select Seats

How to Join

1. Get the exact event name and 6-character room code from the event organizer.
2. Enter the event name exactly as it was created (case-sensitive)
3. Enter the room code (automatically converted to uppercase)
4. Click "Join Event" - you'll be added if the event isn't full

Important Notes

- Both event name and room code must match exactly
- You cannot join an event that's already at full capacity
- You can only join an event once
- Event organizers can see all participants

Chat

VISUAL REPRESENTATION



Browser tabs: Profile | Wintel | Authentication Successful | Windows | New tab | Smart Campus Booking System

Address bar: localhost:3000/create-event

Smart Campus | Home | Dashboard | Booking | Create Event | Join Event | Admin Panel | Abhishek | Sign out

Create Event

Organize an event and share the room code with participants

Event Name

Description

Start Date & Time

End Date & Time

Maximum Capacity

Create Event

Tips

- A unique 6-character room code will be generated automatically.
- Share the room code with participants to allow them to join.
- Participants will need both the event name and room code to join.
- Events cannot exceed the specified capacity.
- You can manage your events from the dashboard.

Chat icon

Future Scope

- **Mobile Application:** Development of native mobile applications for Android and iOS to provide better accessibility and enable users to book resources and manage events on the go.
- **Advanced Analytics:** Implementation of detailed analytics with graphical insights to help administrators track resource usage, booking trends, and optimize utilization.
- **IoT Integration:** Integration with smart devices such as automated entry systems or smart displays to provide real-time updates on room availability and bookings.
- **Multi-language Support:** Adding support for multiple languages to make the platform accessible to a wider range of users across different regions.
- **Notification System:** Introducing real-time notifications for booking confirmations, status updates, and event reminders.
- **Custom Resource Management:** Allowing administrators to create and configure custom resources, seating layouts, and booking rules based on institutional needs, portals, and student IDs.

- The Smart Campus Booking System successfully addresses the challenges of managing campus resources and event coordination within educational institutions. By integrating real-time seat availability, resource booking, and event management, the system provides an efficient and practical solution for organizing campus activities.
- The platform ensures that users can book resources without conflicts through realtime seat selection and structured booking processes. Its organized approach, combined with role-based access and administrative control, helps maintain transparency and efficient utilization of resources.
- Built using modern web technologies, the system is scalable, efficient, and easy to maintain. The Smart Campus Booking System demonstrates how technology can be effectively utilized to improve resource management and operational efficiency in institutional environments, making it a valuable solution for modern campuses.

React (Official Documentation)

Meta. *React – A JavaScript Library for Building User Interfaces.*

<https://react.dev>

Vite (Official Documentation)

Evan You. *Vite – Next Generation Frontend Tooling.*

<https://vitejs.dev>

Tailwind CSS (Official Documentation)

Tailwind Labs. *Tailwind CSS – Utility-First CSS Framework.*

<https://tailwindcss.com/docs>

MongoDB (Official Documentation): MongoDB Inc. *The Document Database for Modern Applications.*

<https://www.mongodb.com/docs>

Express.js (Official Documentation)

OpenJS Foundation. *Express – Fast, Unopinionated Web Framework for Node.js.*

<https://expressjs.com>

Node.js (Official Documentation): OpenJS Foundation. *Node.js – JavaScript Runtime built on Chrome's V8 JavaScript engine.*

<https://nodejs.org/en/docs>

Thank You